From the Dot Product to Oriented Area

A Geometric Introduction to Determinants and the Wedge Product

Learning goals. Derive the law of cosines using projections and Pythagoras; prove the coordinate dot product; find parallelogram area; justify the determinant's orientation sign; and expand the wedge product with all four terms visible.

Logical route. Vectors → projections → Pythagoras → law of cosines → dot product → area → determinant → orientation → wedge product.

Visual legend. A solid arrow marks $\boldsymbol{u}$ (blue); a dashed arrow marks $\boldsymbol{v}$ (teal). Orange emphasizes a directed projection or the current comparison. A dashed perpendicular marks height. Faint lines retain earlier construction. Names and formulas keep their meaning in grayscale. Coordinate drawings have equal axis scales; term maps are explicitly algebraic schematics.

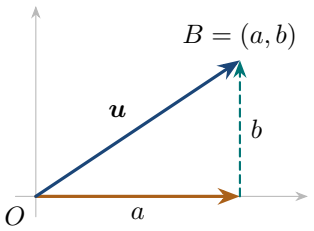
1 Vectors, lengths, and angles

Work in the Euclidean plane $\mathbb{R}^2$ with its standard orthonormal axes:

$$\boldsymbol{e}_1 = (1,0), \quad \boldsymbol{e}_2 = (0,1), \qquad \boldsymbol{u} = (a,b) = a\boldsymbol{e}_1 + b\boldsymbol{e}_2, \quad \boldsymbol{v} = (c,d) = c\boldsymbol{e}_1 + d\boldsymbol{e}_2.$$

Step 1. Read coordinates as displacements

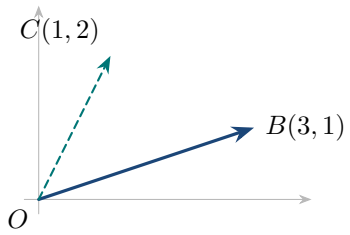
Start at O. Move horizontally by a, then vertically by b. The endpoint is $B = (a,b)$, so the direct arrow OB is $\boldsymbol{u}$. The pictured a, b are positive. A negative coordinate means displacement in the opposite axis direction, not a negative physical length.



Subtraction connects endpoints

Step 2. Draw both arrows from a common origin

Use $\boldsymbol{u} = (3, 1)$ and $\boldsymbol{v} = (1, 2)$. Start both arrows at O and name their tips B and C.
Before subtracting, notice that the two tips are different. We will find the displacement that takes C to B.

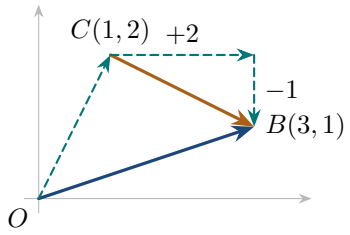


Step 3. Try a numerical displacement first

Draw $\boldsymbol{u} = (3, 1)$ and $\boldsymbol{v} = (1, 2)$ from O. To go from $C = (1, 2)$ to $B = (3, 1)$, move 2 units right and 1 down:

$$\overrightarrow{CB} = (2, -1) = (3 - 1, 1 - 2).$$

This endpoint displacement is $\boldsymbol{u} - \boldsymbol{v}$.

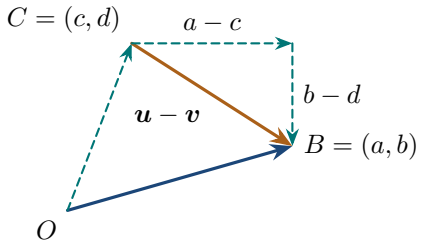


Step 4. Connect the general endpoints

Keep the common tail O. Name the tips $B = (a, b)$ and $C = (c, d)$. Following the route $C \to O \to B$ gives

$$\overrightarrow{CB} = -\boldsymbol{v} + \boldsymbol{u} = \boldsymbol{u} - \boldsymbol{v}.$$

Equivalently, $\boldsymbol{v} + \overrightarrow{CB} = \boldsymbol{u}$. The dashed displacement route ends at the same B.



Lengths from Pythagoras. Signed displacements become nonnegative squared lengths:

$$\boldsymbol{u} - \boldsymbol{v} = (a - c, b - d), \qquad \|\boldsymbol{u}\| = \sqrt{a^2 + b^2}, \quad \|\boldsymbol{v}\| = \sqrt{c^2 + d^2},$$

$$\|\boldsymbol{u} - \boldsymbol{v}\|^2 = (a - c)^2 + (b - d)^2.$$

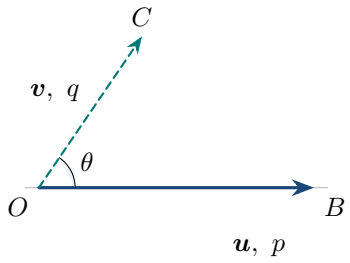
For nonzero vectors, $\theta \in [0, \pi]$ is their *unoriented* included angle. A zero vector has no uniquely defined angle with another vector.

2 Projections and elementary trigonometry

Assume both vectors are nonzero. Write $p = \|\boldsymbol{u}\|$ and $q = \|\boldsymbol{v}\|$. Rotate the *drawing* so OB points right; reflect the view if necessary so C is above the line for this unoriented calculation. This preserves lengths and θ. It does not assert that the original coordinates (a, b) have become $(p, 0)$. We will return to the original coordinate axes later.

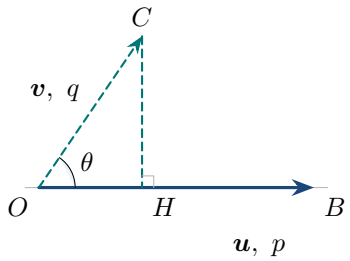
Step 1. Draw the two vectors

Name the common tail O, the tip of $\boldsymbol{u}$ by B, and the tip of $\boldsymbol{v}$ by C. The angle $\theta = \angle BOC$ lies between sides of lengths p and q.
First identify these sides. No dot product is being used.



Step 2. Drop a perpendicular to the whole line

Let H be the foot of the perpendicular from C to the line OB. The right-angle mark says $CH \perp OB$.
Use the whole line, because H need not lie between O and B. This construction creates the right triangle needed for trigonometry.

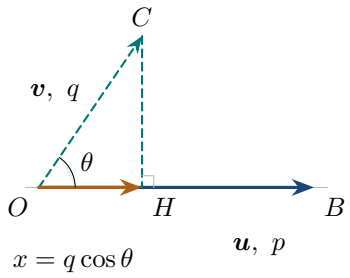


Step 3. Name the signed shadow

Let x be the directed coordinate of H, measured from O toward B:

$$x = q\cos\theta.$$

In the pictured acute triangle, $\cos\theta = OH/q$, so $x = OH > 0$. This scalar is the signed component $\mathrm{comp}_{\boldsymbol{u}}(\boldsymbol{v})$.



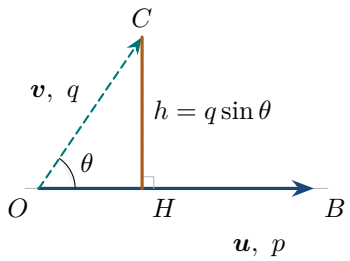
A height is a distance; a projection may be signed

Step 4. Name the nonnegative height

The physical distance $CH = h$ is perpendicular to the base. In the pictured acute triangle, $\sin\theta = h/q$, hence

$$h = q\sin\theta \geq 0.$$

For $\theta \in [0, \pi]$, sine remains nonnegative. The sloping side has length q; it is not the height.

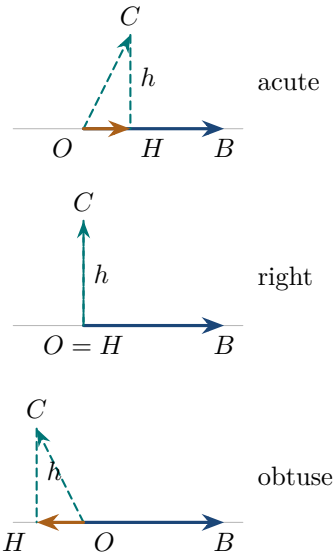


Step 5. Compare the three angle types

The foot H can move across O. These panels use the same $p = 3$, $q = \sqrt{5}$ and equal scales.
Acute: $x = 1$, $h = 2$.
Right: $x = 0$, $h = \sqrt{5}$.
Obtuse: $x = -1$, $h = 2$.
For the obtuse case, the acute reference angle is $\pi - \theta$. Thus

$$OH = q\cos(\pi - \theta) = -q\cos\theta = |x|.$$

The signed x is negative while the length OH is positive.



Pythagorean trigonometric identity. The legs of OHC have lengths $|x|$ and h, so $x^2 + h^2 = q^2$. Divide by $q^2 > 0$:

$$(x/q)^2 + (h/q)^2 = 1 \quad \Longrightarrow \quad \cos^2\theta + \sin^2\theta = 1.$$

At 90°, $H = O$ and the identity holds directly. At 0 or π, the height is zero and the same formulas hold by the collinear geometry.

Pause and predict. If H moves left of O, does the physical height become negative?

Explanation. No. Only the along-base projection x changes sign. Here h is a distance. A signed perpendicular height will be introduced with orientation.

3 Pythagoras and the law of cosines

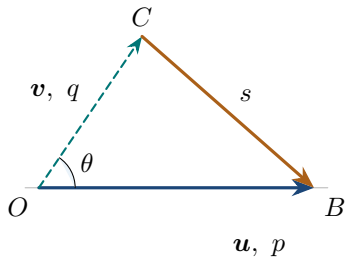
Goal. Find the endpoint distance $s = BC = \|\boldsymbol{u} - \boldsymbol{v}\|$ from p, q, and θ, without assuming a dot-product formula.

Step 1. Identify the opposite side

The vector difference points $C \to B$, and its length is s. The side BC is opposite θ.

$$OB = p, \quad OC = q, \quad BC = s.$$

Keep the foot H from the preceding construction. Its directed coordinate is $x = q\cos\theta$.

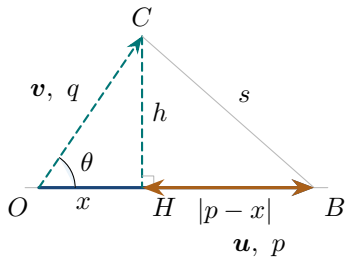


Step 2. Read the horizontal separation

Relative to the rotated baseline, B has coordinate p and H has coordinate x. The directed difference from H to B is $p - x$.

$$HB = |p - x|, \qquad CH = h.$$

Here $0 < x < p$, so $HB = p - x$. The absolute-value form remains valid outside this configuration.

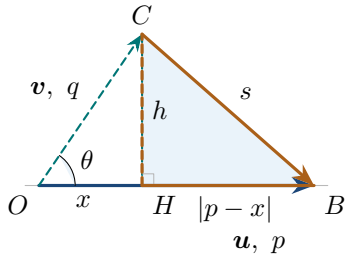


Step 3. Apply Pythagoras to the highlighted triangle

In the right triangle BHC, the hypotenuse is s. Therefore

$$s^2 = |p - x|^2 + h^2 = (p - x)^2 + h^2.$$

Squaring removes the sign of the directed horizontal difference. It does not make a negative difference a physical length.



Expand algebra, then use the second right triangle

Step 4. Expose both cross terms

Distribute each factor of $(p-x)(p-x)$:

$$\begin{aligned}(p-x)^2 &= p^2 - px - xp + x^2 \\ &= p^2 - 2px + x^2.\end{aligned}$$

There are two cross terms, one from each choice of a p and an $-x$. Commutativity gives $xp = px$.
This is a term map, not a dissection into physical areas; x may be negative.

Algebraic term map

$\times$	p	$-x$
p	p^2	$-px$
$-x$	$-xp$	x^2

Each row distributes one term. Read all four cells before collecting.

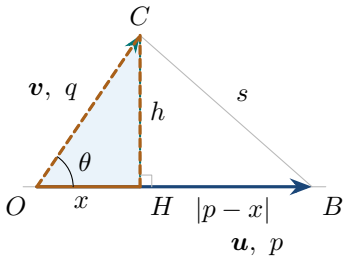
Step 5. Replace the two squared legs by the hypotenuse

Now highlight OHC. Its legs have lengths $|x|$ and h, so

$$x^2 + h^2 = q^2.$$

Substitute this exact sum:

$$\begin{aligned}s^2 &= p^2 - 2px + (x^2 + h^2) \\ &= p^2 - 2px + q^2.\end{aligned}$$

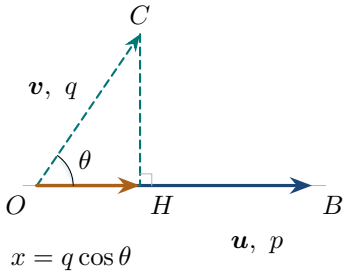


Step 6. Insert the signed projection

Replace x by $q\cos\theta$:

$$\begin{aligned}s^2 &= p^2 + q^2 - 2p(q\cos\theta) \\ &= p^2 + q^2 - 2pq\cos\theta.\end{aligned}$$

The cosine appears because the endpoint distance depends on the along-base shadow of OC.



Theorem (law of cosines). For sides $p, q > 0$ including angle θ and opposite side s,

$$s^2 = p^2 + q^2 - 2pq\cos\theta.$$

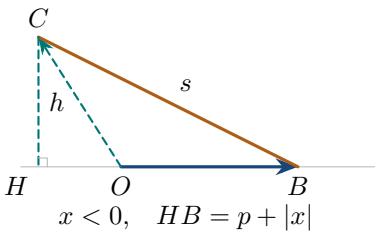
Check the foot outside the base, then check a right angle

Step 7. Let the projection be negative

For an obtuse angle, H lies behind O. Write $x = -|x|$. Then

$$\begin{aligned} s^2 &= (p + |x|)^2 + h^2 \\ &= p^2 + 2p|x| + (x^2 + h^2) \\ &= p^2 + q^2 - 2px. \end{aligned}$$

Since $x = q\cos\theta < 0$, the correction $-2pq\cos\theta$ is positive. The opposite side is longer than $\sqrt{p^2 + q^2}$.



Step 8. Let the foot lie beyond B

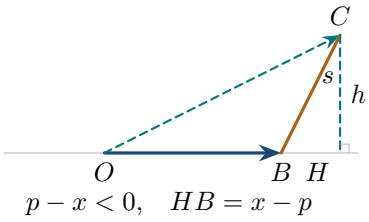
For example, use $p = 3$, $x = 4$, $h = 2$. The angle is acute, but H lies beyond B:

$$p - x = -1, \qquad HB = |p - x| = 1.$$

Then $s^2 = (-1)^2 + 2^2 = 5$. Also $q^2 = 4^2 + 2^2 = 20$, so

$$p^2 + q^2 - 2px = 9 + 20 - 24 = 5.$$

The sign of $p - x$ does not affect its square.

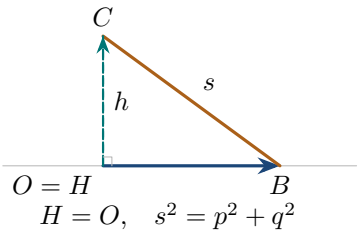


Step 9. End with the right-angle check

At $\theta = 90°$, $x = q\cos\theta = 0$ and $H = O$. The law reduces to

$$s^2 = p^2 + q^2.$$

This is Pythagoras in triangle OBC. The collinear limits $\theta = 0, \pi$ give $s = |p - q|$ and $s = p + q$, respectively.



4 The dot product: geometry and coordinates

Definition. For nonzero vectors, define the geometric dot product by

$$\boldsymbol{u} \cdot \boldsymbol{v} := \|\boldsymbol{u}\| \, \|\boldsymbol{v}\| \cos\theta = \|\boldsymbol{u}\| \operatorname{comp}_{\boldsymbol{u}}(\boldsymbol{v}).$$

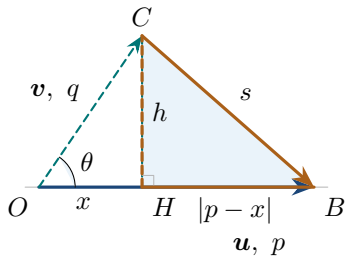
For a pair containing a zero vector, define the dot product to be zero. We now *prove*, rather than assume, its coordinate formula.

Step 1. Calculate the same segment using geometry

The law of cosines concerns $s = BC$:

$$\begin{aligned} s^2 &= p^2 + q^2 - 2pq\cos\theta \\ &= a^2 + b^2 + c^2 + d^2 \\ &\quad - 2\,\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\cos\theta. \end{aligned}$$

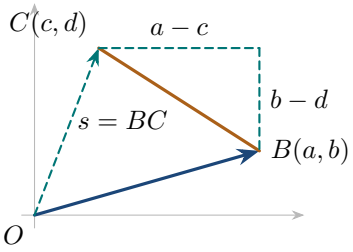
The lengths satisfy $p^2 = a^2 + b^2$ and $q^2 = c^2 + d^2$ in the original orthonormal coordinates.



Step 2. Calculate that segment using coordinates

Return to the original coordinate axes. The same endpoints are $B = (a, b)$ and $C = (c, d)$:

$$\begin{aligned} s^2 &= (a-c)^2 + (b-d)^2 \\ &= (a^2 - ac - ca + c^2) \\ &\quad + (b^2 - bd - db + d^2) \\ &= a^2 + b^2 + c^2 + d^2 - 2(ac + bd). \end{aligned}$$



Each square has two cross terms, as in the preceding term map. No coordinate dot-product formula was used.

Pause and predict. Why can we equate the two expressions despite the different views?

Explanation. Both calculate the squared length of the same segment BC. A rigid change of view preserves that length.

Equate, subtract, and divide

Step 3. Remove the common squared lengths

Abbreviate $S = a^2 + b^2 + c^2 + d^2$ and $G = \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\cos\theta$. Then

$$\begin{aligned} S - 2(ac+bd) &= S - 2G, \\ [S - 2(ac+bd)] - S &= [S - 2G] - S, \\ -2(ac+bd) &= -2G, \\ \frac{-2(ac+bd)}{-2} &= \frac{-2G}{-2}, \\ ac + bd &= G = \boldsymbol{u}\cdot\boldsymbol{v}. \end{aligned}$$

Subtracting the same S preserves equality; dividing by the nonzero scalar -2 preserves it too.

Algebraic comparison

One segment BC
Two formulas for s^2
$\downarrow$ subtract S
$\downarrow$ divide by -2
One coordinate formula

$$\boldsymbol{u}\cdot\boldsymbol{v} = ac + bd.$$

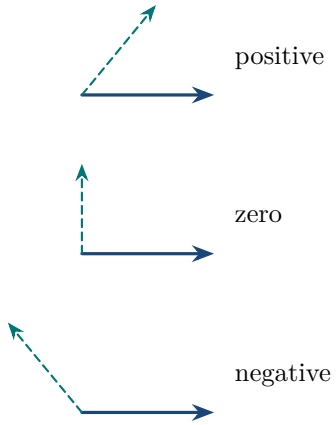
If $\boldsymbol{u} = \boldsymbol{0}$, then $a = b = 0$ and $ac + bd = 0$. The same argument applies if $\boldsymbol{v} = \boldsymbol{0}$, so the theorem holds for every pair.

Step 4. Read the sign as alignment

For nonzero vectors, $pq > 0$. The sign of the dot product equals that of $\cos\theta$:

$\theta < 90^\circ$	$\boldsymbol{u}\cdot\boldsymbol{v} > 0$
$\theta = 90^\circ$	$\boldsymbol{u}\cdot\boldsymbol{v} = 0$
$\theta > 90^\circ$	$\boldsymbol{u}\cdot\boldsymbol{v} < 0$

For non-collinear pairs these are the acute, perpendicular, and obtuse cases. At the collinear endpoints, $\theta = 0$ gives positive and $\theta = \pi$ negative alignment.



Useful identities follow from the proved formula

Length and symmetry.

$$\boldsymbol{u}\cdot\boldsymbol{u} = a\cdot a + b\cdot b = a^2 + b^2 = \|\boldsymbol{u}\|^2, \qquad \boldsymbol{v}\cdot\boldsymbol{u} = ca + db = ac + bd = \boldsymbol{u}\cdot\boldsymbol{v}.$$

These identities use ordinary scalar multiplication of coordinates.

Distributivity. Let $\boldsymbol{w} = (e, f)$. Distribute in coordinates and regroup:

$$\begin{aligned}(\boldsymbol{u}+\boldsymbol{w})\cdot\boldsymbol{v} &= (a+e)c + (b+f)d \\ &= ac + ec + bd + fd \\ &= (ac+bd) + (ec+fd) \\ &= \boldsymbol{u}\cdot\boldsymbol{v} + \boldsymbol{w}\cdot\boldsymbol{v}.\end{aligned}$$

Similarly, for any scalar t,

$$(t\boldsymbol{u})\cdot\boldsymbol{v} = (ta)c + (tb)d = t(ac+bd) = t(\boldsymbol{u}\cdot\boldsymbol{v}).$$

Symmetry gives linearity in the other input as well. This is bilinearity.

Algebraic grouping schematic (not a geometric area diagram).

$$\underbrace{ac+bd}_{\boldsymbol{u}\cdot\boldsymbol{v}} \quad + \quad \underbrace{ec+fd}_{\boldsymbol{w}\cdot\boldsymbol{v}}.$$

The terms are grouped by which vector supplied the first coordinate factor.

Pause and predict. Can the dot product alone tell clockwise from counterclockwise?

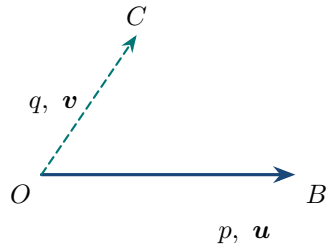
Explanation. No. Reflecting a pair reverses its orientation while keeping its lengths and unoriented angle. The cosine, and hence the dot product, stays the same. We need a perpendicular measurement with a sign convention.

5 Base times height: the parallelogram area

We first find ordinary, nonnegative area A. Orientation will come later.

Step 1. Start with the two adjacent sides

Place $\boldsymbol{u}$ and $\boldsymbol{v}$ tail-to-tail. The segment OB will be the base, of length $p = \|\boldsymbol{u}\|$. The sloping side OC has length $q = \|\boldsymbol{v}\|$.
This sequence keeps the same O, B, C throughout.

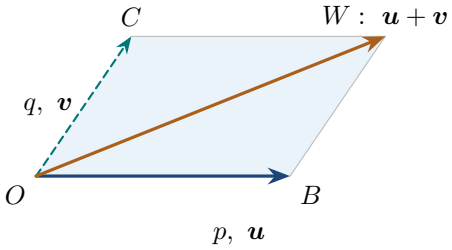


Step 2. Complete by vector addition

Translate $\boldsymbol{v}$ to start at B, and translate $\boldsymbol{u}$ to start at C. Their tips coincide at W, because

$$\overrightarrow{OW} = \boldsymbol{u} + \boldsymbol{v}.$$

Thus $OBWC$ is a parallelogram. Opposite sides are parallel and equal.

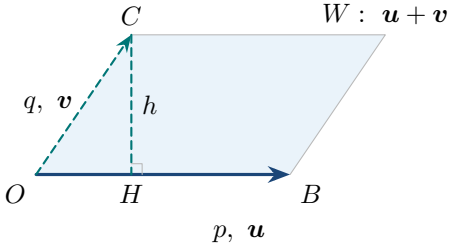


Step 3. Measure perpendicular height

Drop $CH \perp OB$. The distance between the parallel lines OB and CW is

$$h = CH = q \sin\theta.$$

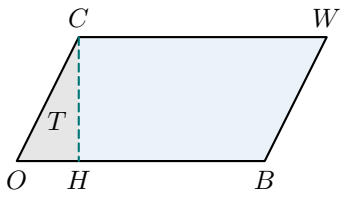
The dashed height is shorter than the sloping side unless the angle is right. Area depends on this perpendicular distance.



Why base times height gives the area

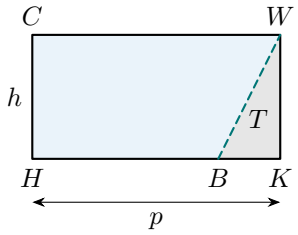
Step 4. Cut off one right triangle

Use the pictured configuration $O = (0, 0)$, $B = (4, 0)$, $C = (1, 2)$, $W = (5, 2)$ and $H = (1, 0)$. Here $0 < OH < OB$, so the cut CH lies inside the parallelogram.
Cut along CH. The triangle OHC, marked T, separates from the remaining trapezoid $HBWC$.



Step 5. Translate the matching piece

Move T by $\boldsymbol{u} = (4, 0)$. Then $O \mapsto B$, $H \mapsto K = (5, 0)$, and $C \mapsto W$. The translated triangle is BKW, also marked T. Translation preserves area. The two pieces fill rectangle $HKWC$ with width $HK = OB = p$ and height h. Hence $A = ph$.



Why the conclusion extends. A left-slanting parallelogram can be reflected to use the same dissection. If the horizontal offset exceeds a base length, cut the figure into finitely many horizontal strips so each strip's offset is smaller than p. Each strip has base length p and the demonstrated area p times its height. Adding the strip areas gives p times the total height. Zero height gives zero area.

$$A = ph = p(q \sin\theta) = \|\boldsymbol{u}\| \, \|\boldsymbol{v}\| \sin\theta \qquad (0 \le \theta \le \pi).$$

If either vector is zero, the figure collapses and $A = 0$. For a directed angle φ, the unsigned formula uses $|\sin\varphi|$.

6 Deriving the determinant formula from the area identity

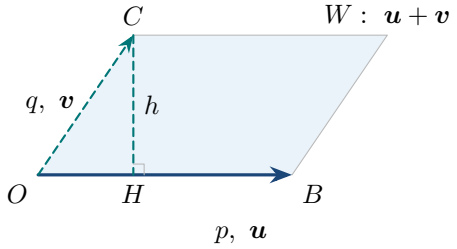
Goal. Eliminate the angle from the area formula. Here A remains unsigned.

Step 1. Square the geometric formula

Start with $A = pq\sin\theta$. Squaring gives

$$A^2 = p^2q^2\sin^2\theta.$$

The square records the size, but cannot retain an orientation sign. That sign has not yet been assigned.

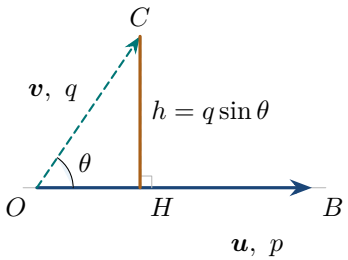


Step 2. Replace sine squared

Pythagoras gave $\sin^2\theta = 1 - \cos^2\theta$. Therefore

$$\begin{aligned} A^2 &= p^2q^2(1 - \cos^2\theta) \\ &= p^2q^2 - p^2q^2\cos^2\theta. \end{aligned}$$

This uses the complementary along-base and perpendicular components of the same $\boldsymbol{v}$.



Step 3. Recognize the squared dot product

Since $pq\cos\theta = \boldsymbol{u}\cdot\boldsymbol{v}$,

$$\begin{aligned} p^2q^2\cos^2\theta &= (pq\cos\theta)^2, \\ A^2 &= p^2q^2 - (\boldsymbol{u}\cdot\boldsymbol{v})^2. \end{aligned}$$

Now substitute the proved coordinate formulas:

$$A^2 = (a^2+b^2)(c^2+d^2) - (ac+bd)^2.$$

Algebraic substitutions

$$p^2 \longmapsto a^2+b^2$$

$$q^2 \longmapsto c^2+d^2$$

$$pq\cos\theta \longmapsto ac+bd$$

Each replacement is an identity already proved.

Expand, cancel, factor, and take the nonnegative root

Step 4. Keep all expansion terms visible

Distribute both products:

$$\begin{aligned}(a^2+b^2)(c^2+d^2) &= a^2c^2 + a^2d^2 \\ &\quad + b^2c^2 + b^2d^2, \\ (ac+bd)^2 &= a^2c^2 + abcd \\ &\quad + abcd + b^2d^2.\end{aligned}$$

Subtract the entire second row. Both $abcd$ terms receive a minus sign.

Algebraic term map

product	subtract square	result
a^2c^2	$-a^2c^2$	0
b^2d^2	$-b^2d^2$	0
a^2d^2	—	a^2d^2
b^2c^2	—	b^2c^2
—	$-abcd - abcd$	$-2abcd$

These fourth-degree terms are algebraic quantities, not literal areas in the original plane.

Step 5. Cancel matching terms and factor

The two matching pairs cancel. The remaining terms form a perfect square:

$$\begin{aligned}A^2 &= a^2d^2 - 2abcd + b^2c^2 \\ &= (ad)^2 - 2(ad)(bc) + (bc)^2 \\ &= (ad-bc)(ad-bc) \\ &= (ad-bc)^2.\end{aligned}$$

Use $r^2 - 2rt + t^2 = (r-t)^2$, with $r = ad$ and $t = bc$.

Factoring schematic

$\times$	ad	$-bc$
ad	$(ad)^2$	$-abcd$
$-bc$	$-abcd$	$(bc)^2$

Reading the four cells backwards reconstructs the factors.

Step 6. Select the nonnegative square root

Physical area satisfies $A \geq 0$. Thus

$$A = \sqrt{A^2} = \sqrt{(ad-bc)^2} = |ad-bc|.$$

Both t and $-t$ square to t^2; its nonnegative root is $|t|$. For instance $\sqrt{(-6)^2} = 6$, not -6.

Unsigned area
$A = |ad-bc| \geq 0$

Signed expression
$ad-bc$ may be
positive, zero, or negative.

Definition. With vectors as columns,

$$D(\boldsymbol{u},\boldsymbol{v}) := \det\begin{pmatrix} a & c \\ b & d \end{pmatrix} := ad-bc, \qquad A = |D(\boldsymbol{u},\boldsymbol{v})|.$$

The expression is $ad-bc$, not $ac-bd$. Transposing gives the same determinant, provided entries are placed consistently. If a vector is zero, both sides of the area identity are zero too.

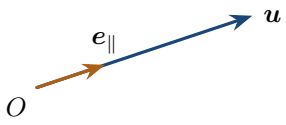
7 Signed area, orientation, and the meaning of subtraction

Orientation convention. The ordered basis $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ is positive. A non-collinear ordered pair is positive when the smaller directed turn from $\boldsymbol{u}$ to $\boldsymbol{v}$ is counterclockwise, and negative when it is clockwise. We justify this sign by signed height, independently of the squared-area identity.

Step 1. Normalize the base direction

For $\boldsymbol{u} = (a, b) \neq \mathbf{0}$, put $p = \sqrt{a^2 + b^2} > 0$. Divide by p:

$$\boldsymbol{e}_{\parallel} = \frac{\boldsymbol{u}}{p} = \left(\frac{a}{p}, \frac{b}{p}\right).$$

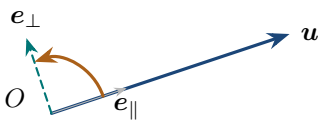


The new arrow points along $\boldsymbol{u}$ but has length 1.

Step 2. Turn the unit direction counterclockwise

A 90° counterclockwise rotation sends (r, s) to $(-s, r)$. Hence

$$\boldsymbol{e}_{\perp} = \left(-\frac{b}{p}, \frac{a}{p}\right).$$



Check both requirements:

$$\boldsymbol{e}_{\parallel} \cdot \boldsymbol{e}_{\perp} = -ab/p^2 + ab/p^2 = 0,$$
$$\|\boldsymbol{e}_{\perp}\|^2 = (b^2 + a^2)/p^2 = 1.$$

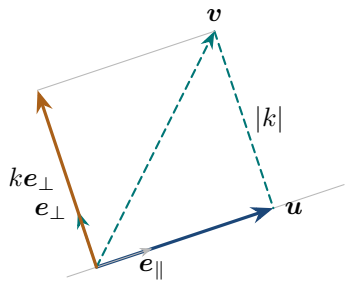
This chooses a particular perpendicular side: the positive side of the ordered base.

Project onto the positive perpendicular

Step 3. Read and calculate signed height

Let $k = \boldsymbol{v} \cdot \boldsymbol{e}_\perp$. This projection is positive on the $\boldsymbol{e}_\perp$ side and negative on the opposite side:

$$\begin{aligned} k &= (c,d) \cdot \left(-\frac{b}{p}, \frac{a}{p}\right) \\ &= -\frac{bc}{p} + \frac{ad}{p} = \frac{ad-bc}{p}. \end{aligned}$$



The physical height is $h = |k|$. The vector projection is $k\boldsymbol{e}_\perp$, whereas k itself is a scalar.

Step 4. Multiply by the positive base length

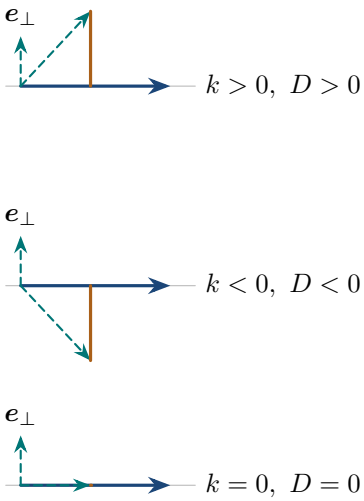
Multiply by $p > 0$:

$$pk = p\frac{ad-bc}{p} = ad - bc = D(\boldsymbol{u}, \boldsymbol{v}).$$

Thus D is base length times *signed* height; $A = p|k| = |D|$ is unsigned.
If $\varphi \in (-\pi, \pi]$ is the directed angle from $\boldsymbol{u}$ to $\boldsymbol{v}$, then

$$k = q \sin\varphi, \qquad D = pq \sin\varphi.$$

Unlike $\theta \in [0, \pi]$, φ keeps the turning direction.



Pause and predict. Could $A^2 = D^2$ alone have told us which side is positive?

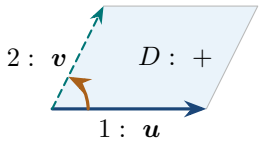
Explanation. No. The counterclockwise choice of $\boldsymbol{e}_\perp$ supplies the convention; the projection calculation proves that $ad - bc$ obeys it.

Order changes sign; dependence collapses area

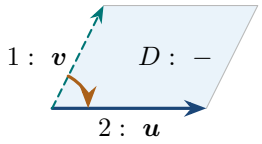
Step 5. Swap the ordered arrows on the same figure

The parallelogram is unchanged. Only the order of the two starting arrows is reversed:

$$\begin{aligned} D(\boldsymbol{v}, \boldsymbol{u}) &= cb - da \\ &= bc - ad = -(ad - bc) \\ &= -D(\boldsymbol{u}, \boldsymbol{v}). \end{aligned}$$



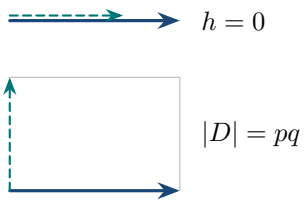
A counterclockwise turn becomes clockwise. The area stays the same because $|-D| = |D|$.



Step 6. Check parallel and perpendicular pairs

If $\boldsymbol{v} = t\boldsymbol{u}$, then $c = ta$, $d = tb$, and

$$D(\boldsymbol{u}, t\boldsymbol{u}) = a(tb) - b(ta) = 0.$$



Conversely, for $\boldsymbol{u} \neq \mathbf{0}$, $D = pk = 0$ forces $k = 0$. Thus $\boldsymbol{v}$ is parallel to $\boldsymbol{u}$.
For a nonzero perpendicular pair, $\boldsymbol{u} \cdot \boldsymbol{v} = 0$, but $|D| = pq > 0$.
If either vector is zero, both products are zero.

8 Three worked examples

Example 1: angle, dot product, and area

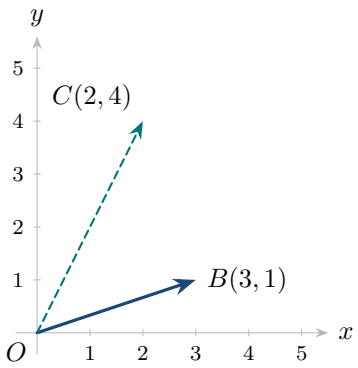
Take $\boldsymbol{u} = (3, 1)$ and $\boldsymbol{v} = (2, 4)$. These panels keep the original coordinate axes and the same unit scale.

Step 1. Place the coordinates and calculate lengths

The tips are $B = (3, 1)$ and $C = (2, 4)$:

$$\begin{aligned} \boldsymbol{u} \cdot \boldsymbol{v} &= 3(2) + 1(4) = 6 + 4 = 10, \\ p &= \sqrt{3^2 + 1^2} = \sqrt{10}, \\ q &= \sqrt{2^2 + 4^2} = \sqrt{20}. \end{aligned}$$

The positive dot product predicts an acute included angle.



Step 2. Find the projection and included angle

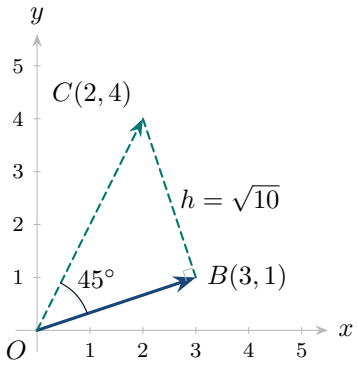
The scalar projection along $\boldsymbol{u}$ is

$$x = \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{p} = \frac{10}{\sqrt{10}} = \sqrt{10} = p.$$

So the perpendicular foot is exactly B. From the geometric definition,

$$\begin{aligned} \cos\theta &= \frac{10}{\sqrt{10}\sqrt{20}} \\ &= \frac{10}{\sqrt{200}} = \frac{1}{\sqrt{2}}, \\ \theta &= 45^\circ. \end{aligned}$$

Indeed $C - B = (-1, 3)$ is perpendicular to $\boldsymbol{u}$, since $3(-1) + 1(3) = 0$.



Example 1 continued: turn the same picture into an area

Step 3. Complete the parallelogram and find height

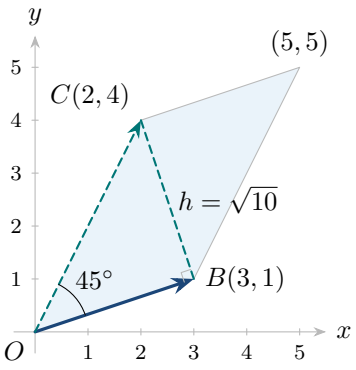
The fourth vertex is $W = B + C = (5, 5)$. The perpendicular height is BC:

$$h = q \sin 45^\circ = \frac{\sqrt{20}}{\sqrt{2}} = \sqrt{10},$$
$$A = ph = \sqrt{10}\sqrt{10} = 10.$$

Equivalently,

$$A = \sqrt{200}\frac{1}{\sqrt{2}} = 10\sqrt{2}\frac{1}{\sqrt{2}} = 10.$$

The dashed BC is a height, not the sloping side OC.

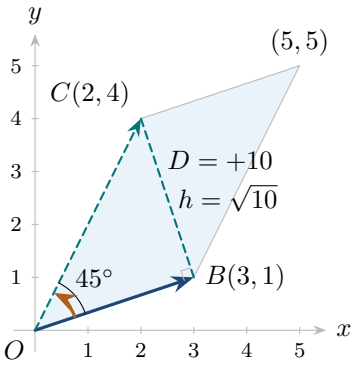


Step 4. Check the determinant and orientation

Calculate in the original coordinates:

$$D(\boldsymbol{u}, \boldsymbol{v}) = 3(4) - 1(2)$$
$$= 12 - 2 = 10,$$
$$A = |10| = 10.$$

The determinant agrees with the geometric area. It is positive because the turn $OB \to OC$ is counterclockwise. The signed height is $k = D/p = \sqrt{10} > 0$, consistent with $\boldsymbol{v}$ lying on the counterclockwise side of $\boldsymbol{u}$.



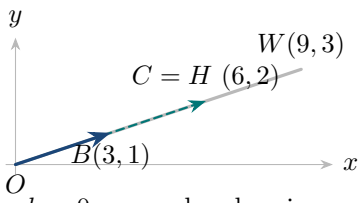
Example 2: parallel vectors

Step 5. See the collapsed parallelogram

Let $\boldsymbol{u} = (3, 1)$ and $\boldsymbol{v} = (6, 2) = 2\boldsymbol{u}$. All four vertices $O, B, C, W = (9, 3)$ lie on one line:

$$\begin{aligned} D(\boldsymbol{u}, \boldsymbol{v}) &= 3(2) - 1(6) \\ &= 6 - 6 = 0, \\ A &= |0| = 0. \end{aligned}$$

The vectors point in the same direction, so $\theta = 0$. The height is $q \sin 0 = 0$; the projected tip H coincides with C.



$h = 0$: no enclosed region

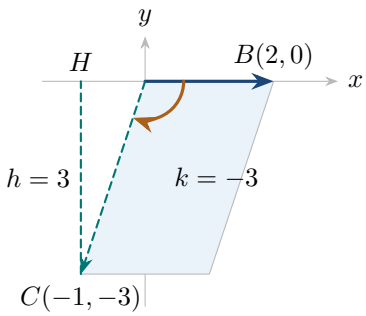
Example 3: an obtuse clockwise pair

Step 6. Separate the two negative signs

Let $\boldsymbol{u} = (2, 0)$ and $\boldsymbol{v} = (-1, -3)$:

$$\begin{aligned} \boldsymbol{u} \cdot \boldsymbol{v} &= 2(-1) + 0(-3) = -2 < 0, \\ D(\boldsymbol{u}, \boldsymbol{v}) &= 2(-3) - 0(-1) = -6 < 0, \\ A &= |-6| = 6. \end{aligned}$$

The along-base projection is $x = -1$: the unoriented angle is obtuse. The signed height is $k = -3$: the directed turn is clockwise. The physical height is 3.
These signs describe different facts. Reflecting C above the horizontal axis reverses D while keeping the dot product negative.



9 An optional next step: the wedge product

Question. Can we keep the oriented area element itself, rather than just its number? The wedge product gives such an object, called a *bivector*.

Definition (exterior algebra in the plane). For $V = \mathbb{R}^2$,

$$\Lambda(V) = \Lambda^0(V) \oplus \Lambda^1(V) \oplus \Lambda^2(V),$$

where $\Lambda^0(V) \cong \mathbb{R}$, $\Lambda^1(V) = V$, and $\Lambda^2(V)$ is one-dimensional with basis $\boldsymbol{e}_1 \wedge \boldsymbol{e}_2$. The product is bilinear and alternating on vectors:

$$\begin{aligned}
(\boldsymbol{x} + \boldsymbol{y}) \wedge \boldsymbol{z} &= \boldsymbol{x} \wedge \boldsymbol{z} + \boldsymbol{y} \wedge \boldsymbol{z}, \\
\boldsymbol{x} \wedge (\boldsymbol{y} + \boldsymbol{z}) &= \boldsymbol{x} \wedge \boldsymbol{y} + \boldsymbol{x} \wedge \boldsymbol{z}, \\
(\lambda \boldsymbol{x}) \wedge \boldsymbol{y} &= \lambda(\boldsymbol{x} \wedge \boldsymbol{y}), \\
\boldsymbol{x} \wedge (\lambda \boldsymbol{y}) &= \lambda(\boldsymbol{x} \wedge \boldsymbol{y}), \\
\boldsymbol{x} \wedge \boldsymbol{x} &= 0.
\end{aligned}$$

These are defining properties, not deductions from the determinant formula.

Step 1. Derive reversal of order. Alternation applied to $\boldsymbol{x} + \boldsymbol{y}$, followed by bilinearity, gives

$$\begin{aligned}
0 &= (\boldsymbol{x} + \boldsymbol{y}) \wedge (\boldsymbol{x} + \boldsymbol{y}) \\
&= \boldsymbol{x} \wedge \boldsymbol{x} + \boldsymbol{x} \wedge \boldsymbol{y} + \boldsymbol{y} \wedge \boldsymbol{x} + \boldsymbol{y} \wedge \boldsymbol{y} \\
&= 0 + \boldsymbol{x} \wedge \boldsymbol{y} + \boldsymbol{y} \wedge \boldsymbol{x} + 0.
\end{aligned}$$

Subtracting $\boldsymbol{x} \wedge \boldsymbol{y}$ gives $\boldsymbol{y} \wedge \boldsymbol{x} = -\boldsymbol{x} \wedge \boldsymbol{y}$.

Step 2. Expand before simplifying. Distribute the left factor, then the right factor, then bring scalars outside:

$$\begin{aligned}
\boldsymbol{u} \wedge \boldsymbol{v} &= (a\boldsymbol{e}_1 + b\boldsymbol{e}_2) \wedge (c\boldsymbol{e}_1 + d\boldsymbol{e}_2) \\
&= (a\boldsymbol{e}_1) \wedge (c\boldsymbol{e}_1 + d\boldsymbol{e}_2) + (b\boldsymbol{e}_2) \wedge (c\boldsymbol{e}_1 + d\boldsymbol{e}_2) \\
&= (a\boldsymbol{e}_1) \wedge (c\boldsymbol{e}_1) + (a\boldsymbol{e}_1) \wedge (d\boldsymbol{e}_2) \\
&\quad + (b\boldsymbol{e}_2) \wedge (c\boldsymbol{e}_1) + (b\boldsymbol{e}_2) \wedge (d\boldsymbol{e}_2) \\
&= ac(\boldsymbol{e}_1 \wedge \boldsymbol{e}_1) + ad(\boldsymbol{e}_1 \wedge \boldsymbol{e}_2) \\
&\quad + bc(\boldsymbol{e}_2 \wedge \boldsymbol{e}_1) + bd(\boldsymbol{e}_2 \wedge \boldsymbol{e}_2).
\end{aligned}$$

Keep all four terms. The next grid explains each basis pair. Coefficient signs remain in the algebra; the grid depicts basis orientations, not scaled component vectors.

Four basis pairs, four terms

$$\begin{aligned}\boldsymbol{u} \wedge \boldsymbol{v} = ac(\boldsymbol{e}_1 \wedge \boldsymbol{e}_1) + ad(\boldsymbol{e}_1 \wedge \boldsymbol{e}_2) \\ + bc(\boldsymbol{e}_2 \wedge \boldsymbol{e}_1) + bd(\boldsymbol{e}_2 \wedge \boldsymbol{e}_2).\end{aligned}$$

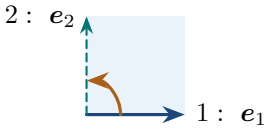
Basis-orientation schematics. Numerals mark the *first* and *second* inputs. Coefficients ac, ad, bc, bd are not represented by arrow lengths.

$ac(\boldsymbol{e}_1 \wedge \boldsymbol{e}_1)$



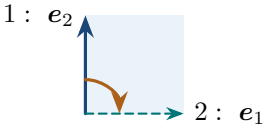
Same direction: collapses to 0

$ad(\boldsymbol{e}_1 \wedge \boldsymbol{e}_2)$



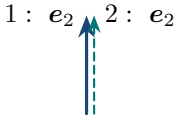
Positive basis orientation

$bc(\boldsymbol{e}_2 \wedge \boldsymbol{e}_1)$



Reversed: $-\boldsymbol{e}_1 \wedge \boldsymbol{e}_2$

$bd(\boldsymbol{e}_2 \wedge \boldsymbol{e}_2)$



Same direction: collapses to 0

Parallel arrows are offset slightly for visibility; they represent a collapsed area, not a thin strip.

Step 3. Substitute the basis relations and collect.

$$\begin{aligned}\boldsymbol{u} \wedge \boldsymbol{v} &= ac(0) + ad(\boldsymbol{e}_1 \wedge \boldsymbol{e}_2) + bc(-\boldsymbol{e}_1 \wedge \boldsymbol{e}_2) + bd(0) \\ &= 0 + ad(\boldsymbol{e}_1 \wedge \boldsymbol{e}_2) - bc(\boldsymbol{e}_1 \wedge \boldsymbol{e}_2) + 0 \\ &= (ad - bc)(\boldsymbol{e}_1 \wedge \boldsymbol{e}_2).\end{aligned}$$

$$\boldsymbol{u} \wedge \boldsymbol{v} = \det \begin{pmatrix} a & c \\ b & d \end{pmatrix} (\boldsymbol{e}_1 \wedge \boldsymbol{e}_2).$$

The bivector is the whole expression. The scalar $ad - bc$ is its coefficient *relative to the chosen oriented basis element* $\boldsymbol{e}_1 \wedge \boldsymbol{e}_2$. Swapping vectors changes its sign. Parallel or zero vectors give the zero bivector.

Optional perspective beyond the plane

In $\mathbb{R}^3$ and higher, wedge products still encode oriented area, but $\Lambda^2(V)$ generally has several independent basis bivectors. Thus the wedge product is not merely another notation for a single plane determinant. The exterior product extends associatively to higher grades. Homogeneous elements satisfy graded commutativity

$$\alpha \wedge \beta = (-1)^{k\ell}\beta \wedge \alpha, \qquad \alpha \in \Lambda^k(V), \quad \beta \in \Lambda^\ell(V).$$

10 Synthesis: two complementary geometric measurements

Question	**Dot product**	**Determinant**
Geometric meaning	Alignment / signed projection	Signed area / orientation
Angle dependence	$\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\cos\theta$	$\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\sin\varphi$
Coordinates	$ac + bd$	$ad - bc$
Zero for nonzero pair	Perpendicular	Parallel
On swapping vectors	Unchanged	Sign reversed

Here $\theta \in [0, \pi]$ is an unoriented angle, whereas φ is directed from $\boldsymbol{u}$ to $\boldsymbol{v}$. A nonzero vector pair can never be simultaneously perpendicular and parallel.

Central idea. The dot product measures how much one vector lies *along* another; the determinant measures its signed *perpendicular* component, multiplied by the base length. The wedge product keeps the corresponding oriented area element as an algebraic object.

Check your understanding

1. For $\boldsymbol{u} = (1, 2)$ and $\boldsymbol{v} = (3, -1)$, compute the dot product and unsigned area. Determine whether the unoriented angle is acute or obtuse, and whether the orientation is positive or negative.
2. Can two nonzero vectors have dot product zero and determinant zero simultaneously? Justify using the geometric meanings, not only coordinates.
3. If $\boldsymbol{v}$ is replaced by $3\boldsymbol{v}$, how do dot product, determinant, and unsigned area change? Repeat for $-\boldsymbol{v}$.
4. Starting from $A^2 = \|\boldsymbol{u}\|^2 \|\boldsymbol{v}\|^2 - (\boldsymbol{u} \cdot \boldsymbol{v})^2$, explain precisely why squaring does not determine orientation.
5. Reproduce the law of cosines for an obtuse angle by writing the signed projection $x < 0$. Identify the point where Pythagoras enters.
6. Expand $(2\boldsymbol{e}_1 - \boldsymbol{e}_2) \wedge (\boldsymbol{e}_1 + 3\boldsymbol{e}_2)$ term by term, then check the coefficient against a 2×2 determinant.